TinyML Compression

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Portability **Trade-offs**

Sacrifice **portability** across systems for **efficiency**.

✗	✓

Specific HW Implementation of a Library

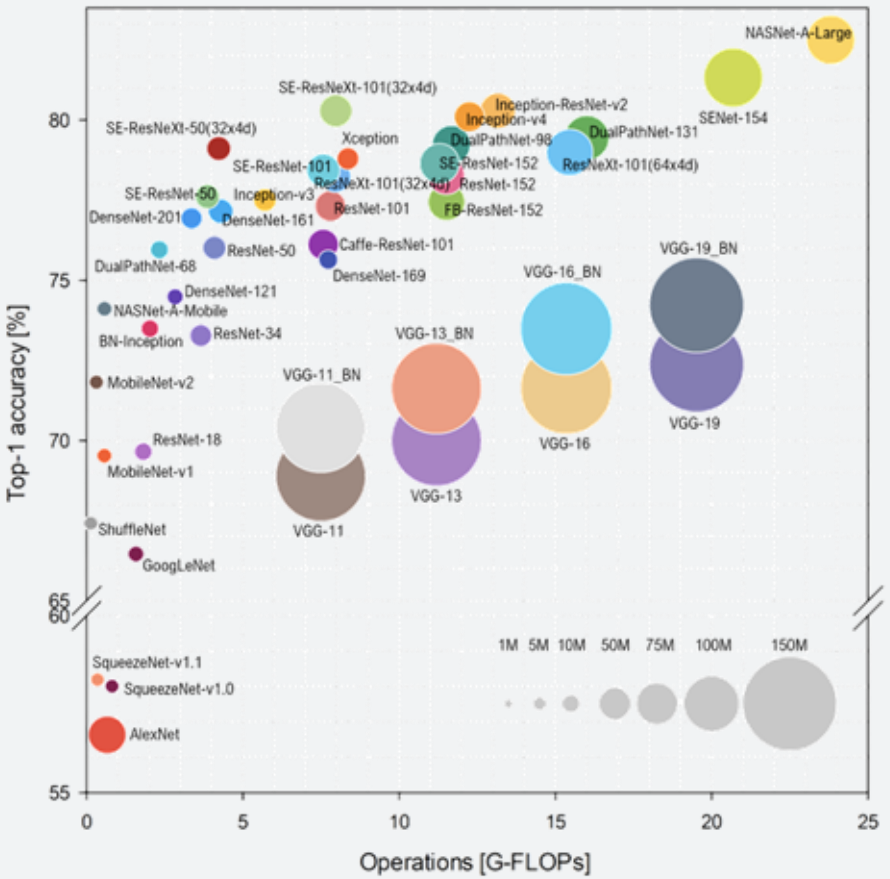
Option 1

- Universal Code Portability/Compatibility ✓
- Cost ($) ✗
- Power Consumption (W) ✗
- Engineering Effort ✗

Option 2

- Lower Code Portability ✗
- Cost ($) ✓
- Power (W) ✓
- Eng. Effort ✓

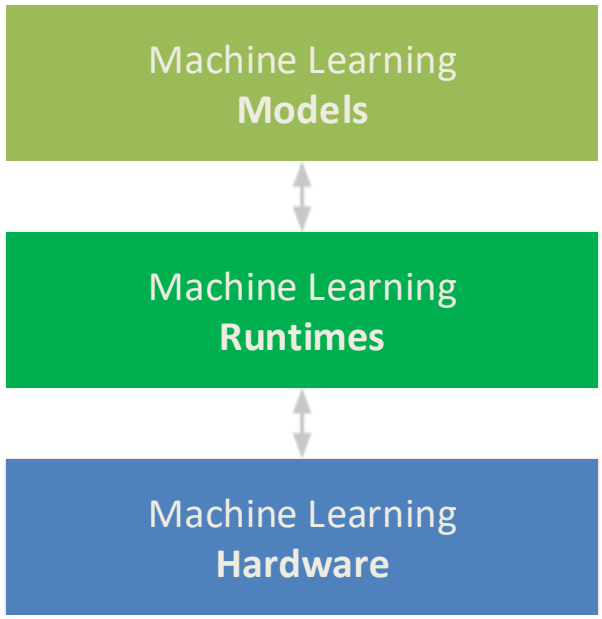
ML Model Evolution



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, "Benchmark analysis of representative deep neural network architectures," *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

Model Comparisons

Model	Version / Variant	Top-1 Accuracy (ImageNet)	Approx. Model Size / Parameters	Comments
AlexNet	AlexNet (2012)	~ 63 % top-1 reported historically. (AI Summer)	~ 60 M parameters, ~ hundreds of MB model size. (Wikipedia)	One of the first deep CNNs, now mostly a teaching tool rather than state-of-the-art.
VGGNet	VGG16 / VGG19	VGG16: ~ 71.3 % top-1 in Keras Applications table. (keras.io)	~ 138 M parameters (VGG16); ~ 528 MB size in Keras table. (keras.io)	Simple, uniform architecture; large size & heavy compute.
ResNet	ResNet152V2 (as a recent version)	ResNet152V2: ~ 78.0 % top-1 (Keras table) (keras.io)	~ 60.4 M parameters (ResNet152/152V2) (keras.io)	Deep residual networks; very good accuracy vs older nets with more efficient size than VGG.
MobileNet	MobileNetV3 (Large)	~ 75.2 % top-1 for MobileNetV3-Large in original paper. (arXiv)	~ 5.48 M parameters for V3-Large (source) (GitHub)	Designed for mobile / edge devices: very efficient size and latency trade-off.



Model Compression Techniques

Pruning

Quantization

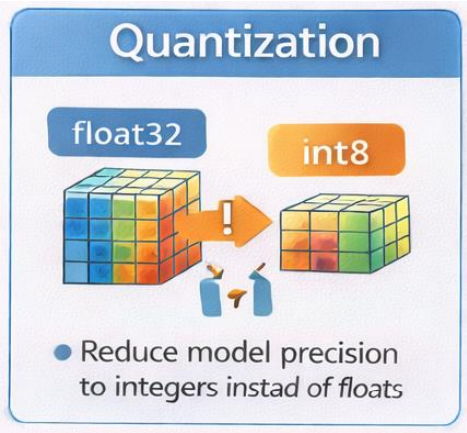
Knowledge

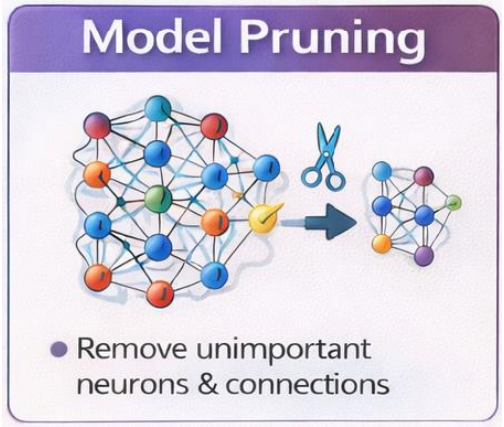
Distillation

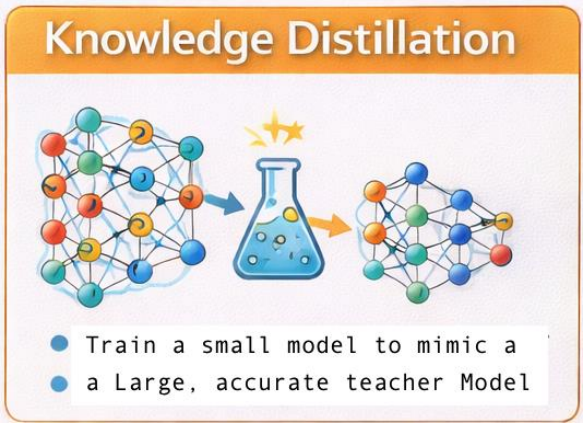
…

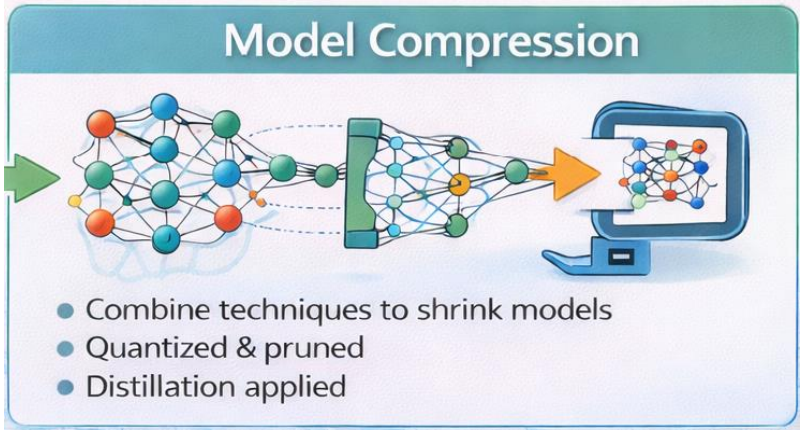
Optimization in Tiny Devices

Embedded ML





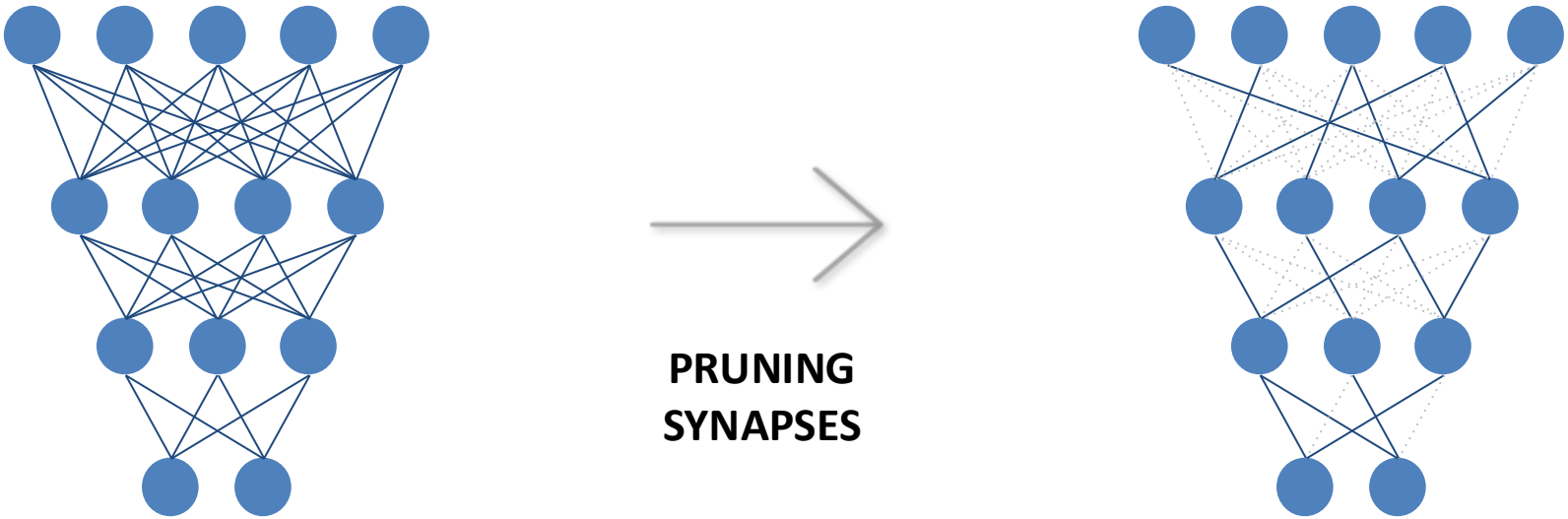




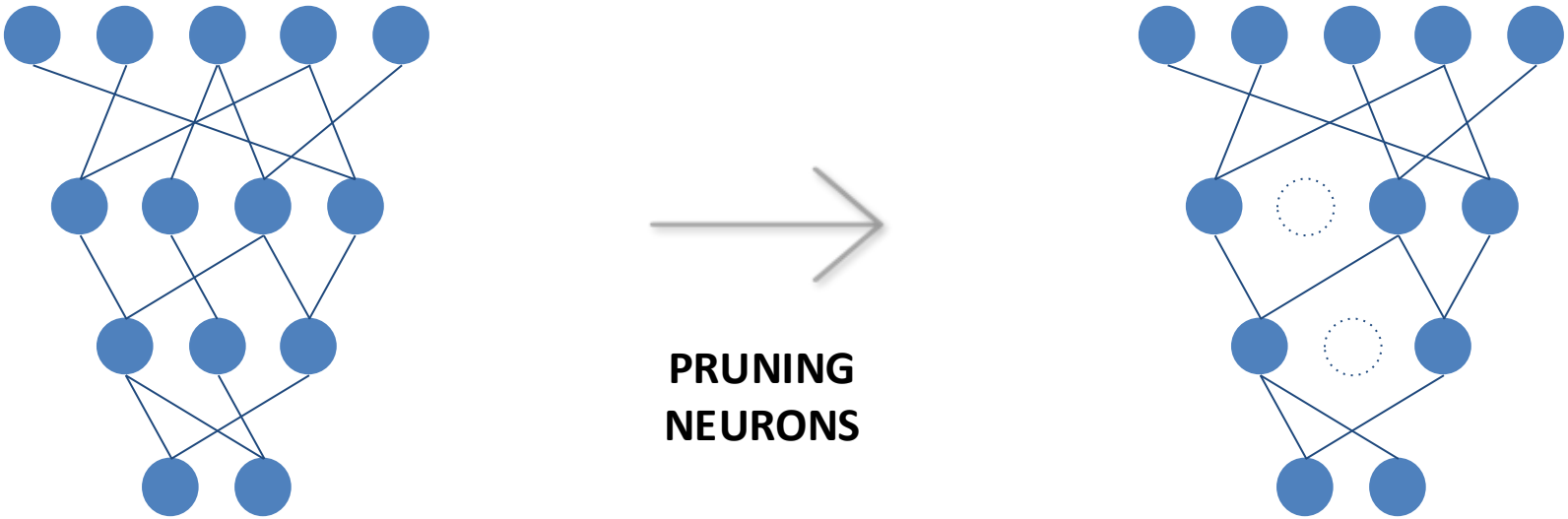
"Smaller models run faster, use less memory, and consume less power"

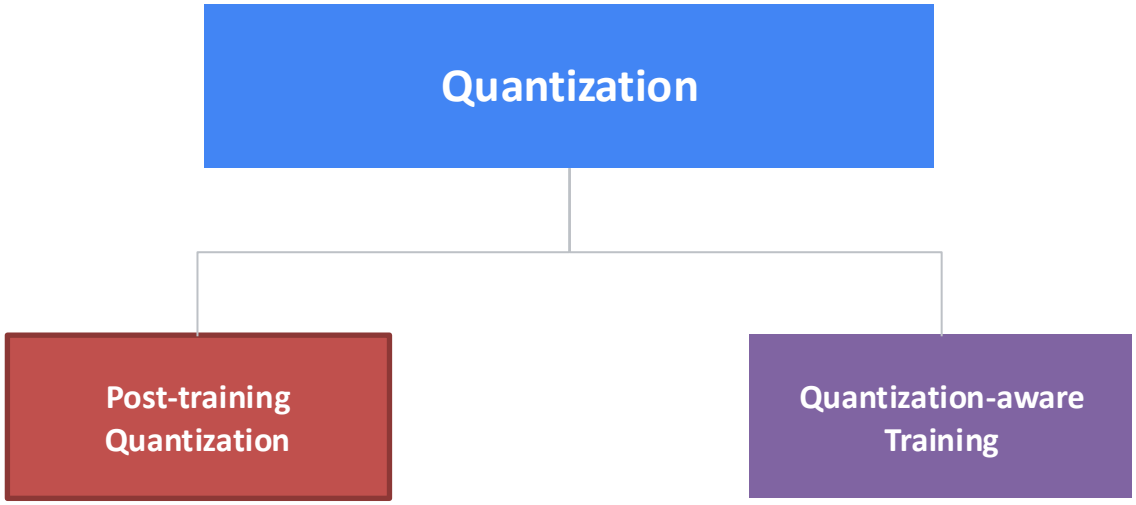
Pruning

Pruning is one model compression technique that allows the model to be optimized for real-time inference for resource-constrained devices.



Pruning



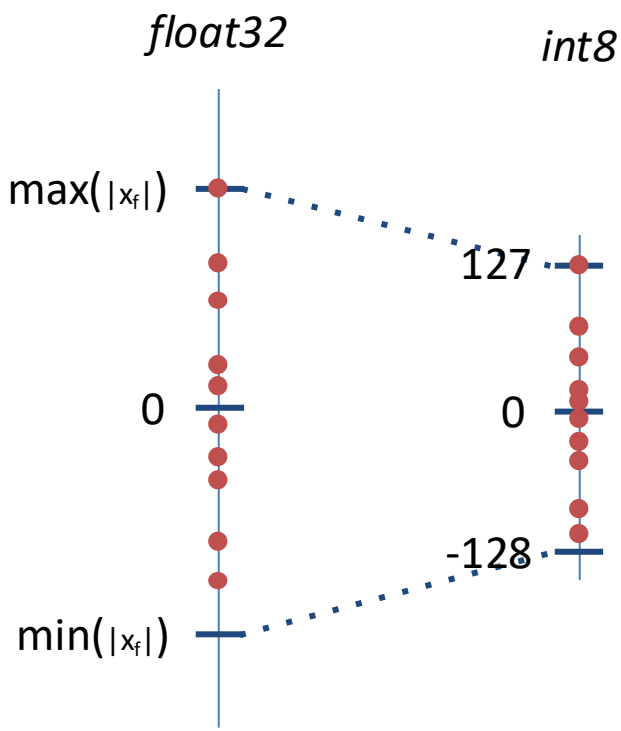


Post-training Optimization Tool (POT) is the fastest and easiest way to get a quantized model. A conversion technique that can reduce model size while also improving CPU and hardware accelerator latency, with little degradation in model accuracy.

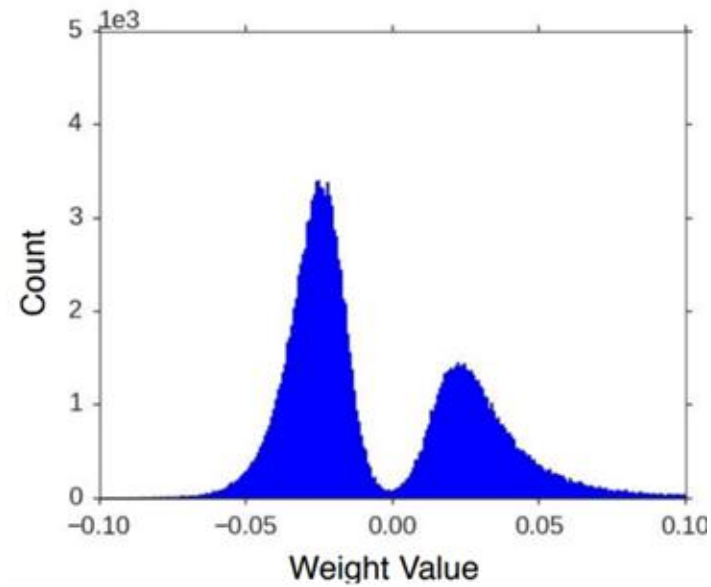
Quantization aware training ***emulates inference-time quantization***, creating a model that downstream tools will use to produce actually quantized models. The quantized models use lower-precision (e.g. 8-bit instead of 32-bit float), leading to benefits during deployment.

Quantization

Quantization is an optimization that works by reducing the precision of the numbers used to represent a model's parameters, which by default are 32-bit floating point numbers. This results in a smaller model size, better portability and faster computation.



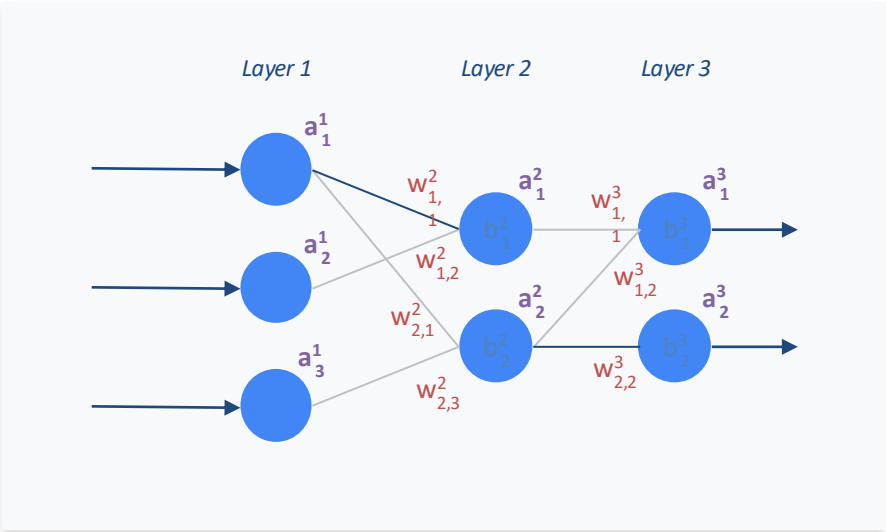
Why it works?
Weight distribution for AlexNet shows how most weight values are concentrated in a small range.



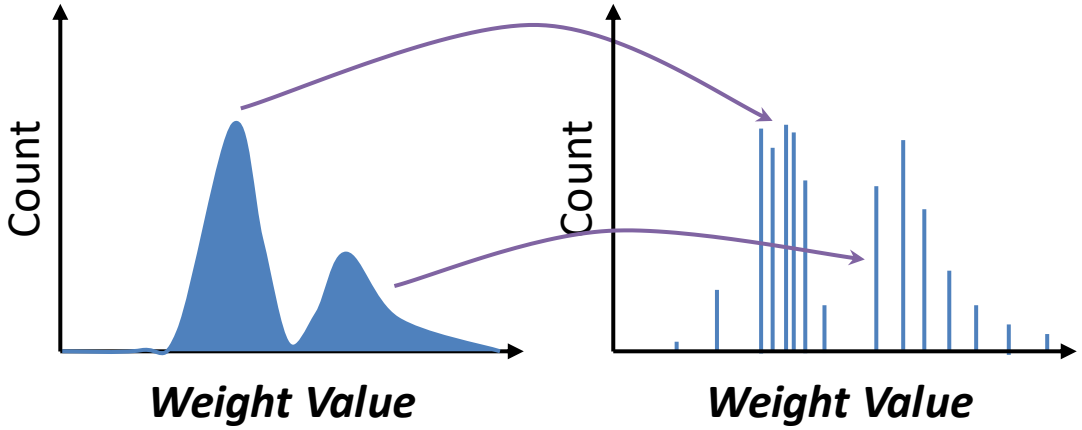
What to Quantize?

Quantize

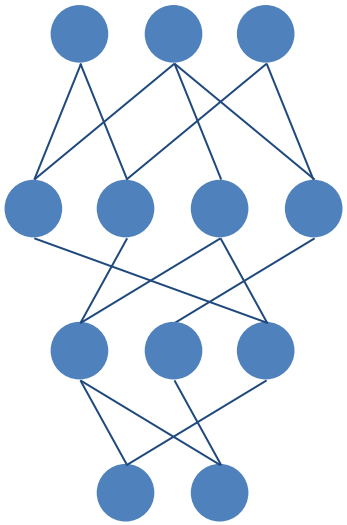
- Weights
- Biases
- Activations

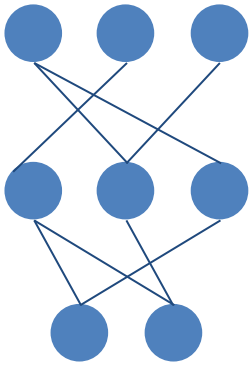


Reduce Precision (Discretize)



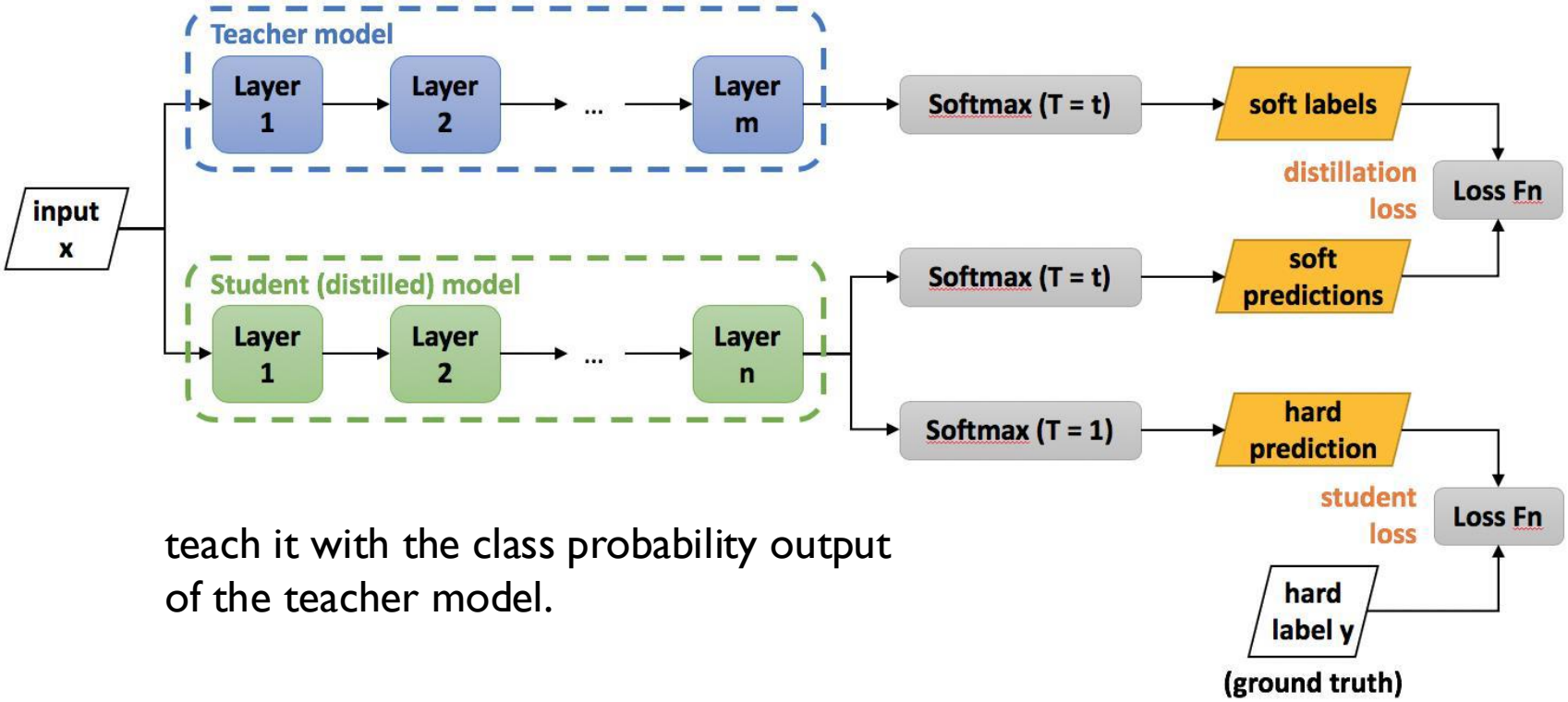
Knowledge Distillation





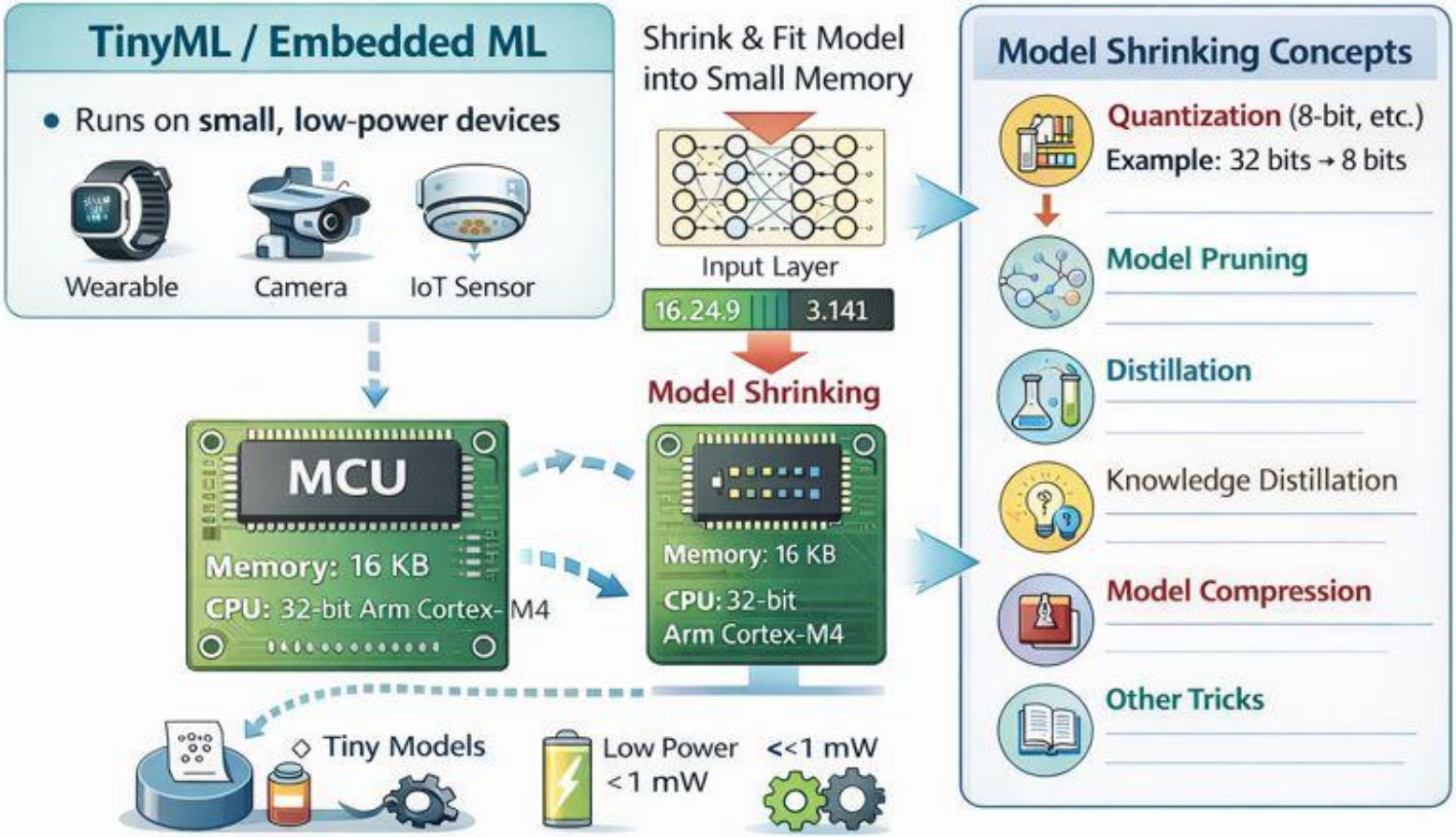
knowledge distillation is the process of transferring knowledge from a large model to a smaller one.

Transferring Knowledge



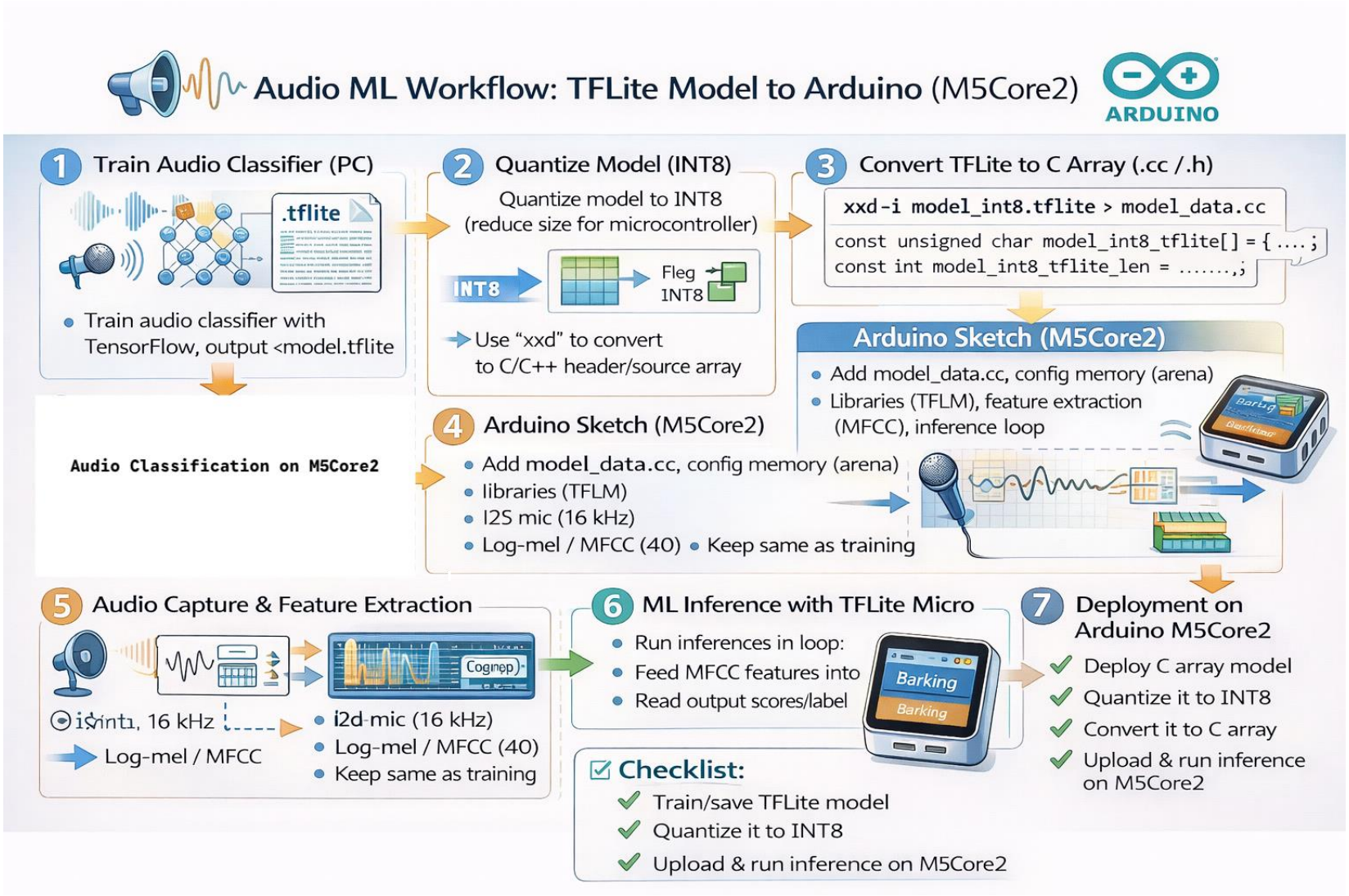
teach it with the class probability output of the teacher model.

TinyML


TinyML / Embedded ML
Runs on small, low-power devices
Wearable
Camera
IoT Sensor
Shrink & Fit Model into Small Memory
Input Layer
16.24.9
3.141
Model Shrinking
MCU
Memory: 16 KB
CPU: 32-bit Arm Cortex-M4
Memory: 16 KB
CPU: 32-bit Arm Cortex-M4
Tiny Models
Low Power
< 1 mW
<<1 mW
Model Shrinking Concepts
Quantization (8-bit, etc.)
Example: 32 bits → 8 bits
Model Pruning
Distillation
Knowledge Distillation
Model Compression
Other Tricks

Deploy in the Tiny Device

Embedded ML



Hands-On

Embedded ML

- Task 1
  - Test capture of imu data (Q) on PC using BLE interface.
  - Test capture of imu data (Q) on PC using WiFi: UDP.
  - Test capture of imu data (Q) on PC using ESP-NOW.

  - Which one is faster? How fast? How does each protocol work?

Hands-On

Embedded ML

- Task II – this is open problem
  - Capture 'action' imu data (Q) on PC using any of the above working protocol.
  - Or you can use the data captured by Serial interface – UART protocol in the previous class.
  - Use CNN to develop the model in "h5" format.
    - Use tensorflow-model-optimization pkg
    - Use pruning when you develop the CNN model
    - If you can't install tensorflow-model-optimization pkg
      - Manual prune the network.
    - What is the model compression %
  - Use TFLiteConverter to optimize by converting float32 to int8
  - Export the model to tflite model, convert the model to C++ header file

Hands-On

Embedded ML

- Compare the accuracy and timing of the models (h5, pruned h5, and quantized tflite)
- Tabulate the results
- If your tflite results are bad – improve them
- Use improved tflite model to perform inference on the M5Core2 board.